

12. The most common carrier of communicable diseases is
 (a) ant (b) housefly
 (c) dragonfly (d) spider
 (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (b)

Houseflies (*Musca domestica*) have long been regarded as potential carriers of microorganisms. It is one of the most common carrier of communicable diseases.

Genetic Engineering and Biotechnology

1. Who synthesized the DNA in vitro?
 (a) Arthur Kornberg (b) Robert Hooke
 (c) Edward Jenner (d) Joseph Lister
 56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

Arthur Kornberg synthesized the DNA in vitro. He won the 1959 Nobel prize in Physiology or Medicine for his discovery of the mechanisms in the biological synthesis of DNA.

2. 'AVAHAN' the Indian segment of 'Bill and Melinda Gates Foundation', is engaged in the prevention of-
 (a) Dengue (b) Polio
 (c) HIV/AIDS (d) Filariasis
 53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

AVAHAN is an initiative sponsored by the Bill & Melinda Gates Foundation to reduce the spread of HIV in India. It began in 2003.

3. Which date is known as 'Diabetes Day'?
 (a) 14th February (b) 14th May
 (c) 14th September (d) 14th November
 53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

World Diabetes Day (WDD) is celebrated every year on November 14.

4. To diagnose the medical conditions, MRI technique is used. Which is not used in MRI?
 (a) Magnetic field (b) X-ray
 (c) Radio wave (d) Gamma ray
 (e) None of the above/More than one of the above
 66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (e)

Magnetic Resonance Imaging (MRI) is a medical imaging technique used in radiology to form pictures of the anatomy and the physiological processes of the body. MRI scanners use strong magnetic fields, magnetic field gradients, and radio waves to generate images of the organs in the body. MRI technique does not involve use of ionizing radiation (Gamma rays, X-rays and the higher ultraviolet part of the electromagnetic spectrum), which distinguishes it from CT and PET scans.

5. The infective stage of *Entamoeba histolytica* is
 (a) trophic form (b) mature cyst
 (c) sporozoite (d) minuta form
 (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

Entamoeba histolytica is an unicellular eukaryotic parasite. *Entamoeba histolytica* causes dysentery by invading the mucosa and submucosa layers of the large intestine in human beings. The infective stage of *Entamoeba histolytica* is mature cyst stage.

Miscellaneous

1. How do vector vaccines work to provide immunity?
 (a) By introducing a weakened or inactivated virus into the body
 (b) By directly attacking and destroying pathogens in the body
 (c) By placing the virus in a modified version of a different virus
 (d) By entering directly into the cells and enabling them to create spike proteins

69th B.P.S.C. (Pre) 2023

Ans. (c)

Vector Vaccines work to provide immunity by placing the virus in a modified version of a different virus. Vector vaccine refers to the introduction of viral gene into a harmless virus.

2. Match List-I with List-II :

List—I (Cloth)	List—II (Origin)
A. Linen	1. Coconut plant
B. Coir	2. Flax plant
C. Mohair	3. Feathers of duck and geese
D. Down	4. Angora goat

 Select the correct answer using the codes given below.

	A	B	C	D
(a)	1	3	2	4
(b)	4	3	1	2
(c)	2	1	4	3
(d)	4	1	3	2

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Ans. (c)

The correct matching is as follows—

List—I (Cloth)	List—II (Origin)
A. Linen	2. Flax plant
B. Coir	1. Coconut plant
C. Mohair	4. Angora goat
D. Down	3. Feathers of duck and geese

3. Which of the following is related to genetic engineering?
 (a) Plastid (b) Plasmid
 (c) Heterosis (d) Mutation
 (e) None of the above / More than one of the above

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Ans. (b)